FAN AND PROPELLER DESIGN VIA INVERSE PROBLEM ADJOINT EQUATIONS

Michele Ferlauto, Angelo Iollo and Luca Zannetti
Dipartimento di Ingegneria Aeronautica e Spaziale
Politecnico di Torino
10129 Torino, ITALY

Abstract

We consider the optimization of a propfan for thrust. An adjoint optimization method based on the solution of an inverse problem is proposed. We study the flow in the meridional plane, and the distributed control is the volume force exerted by the blades on the flow.
The adjoint formulation delivers the thrust gradient with respect to such flow variable distribution, and a descent method can be used for optimization.
As typical applications, the thrust optimization of a ducted and an unducted configuration are presented.

Introduction

Efficient propulsion requires high mass flow rates with moderate velocity increments. In this respect propfans seem an efficient alternative to jet propulsion even at flight speeds above 400 knots.
Yet, commercial applications are rare, because of unsatisfactory aeroelastic blades behavior as well as relevant noise generation.
As a preliminary effort, we consider the optimization of propfan blades for thrust, in the wake of recent papers we dedicated to turbofan design [1].
This is a first step in the direction of tuning a method for noise reduction by blades shape optimization.

While usual optimization methods [2] iterate on the geometry of a certain flow problem to determine a configuration minimizing or maximizing a given objective function, in the present study we propose to iterate on inverse problems [3][4], motivated by the fact that in such formulation fluid-dynamic constraints on the solution are very easily imposed.
This is possible because the controls of the optimization process are in our method flow variables, for example pressure. Hence, the flow constraints can be imposed directly in the parameterization of the controls, with no computational expense. In the design of innovative propfans, indeed, stringent flow constraints must be imposed.
At present, a fully three dimensional propfan design is feasible using at most simplified flow models such as the Euler equations. Therefore, in the optimization stage, boundary layer requirements are taken into account by theoretical or experimental correlations.
This typically results in strict constraints for the load, i.e. the pressure difference between the upper and lower blade surface. With the approach we propose the pressure gradient at the wall can be parameterized in order to obtain mild adverse pressure gradients and attached boundary layers, at no additional computational cost compared to unconstrained optimization methods. Other such problems can be easily treated: when designing a fan it is often required to maximize some performance, given the work per unit mass exerted on the fluid. Or, when several compressor stages are present, it is required that the compression takes place with a given distribution over the stages.
With usual optimization techniques all of these cases would typically be treated by including a penalization term into the performance functional, thus resulting in poor convergence of the optimization. Otherwise, they are accounted by additional constraints to be included in the lagrangian, with additional adjoint equations to be derived and solved.

As concerning geometric constraints, in the present method they are taken into account either by penalization or by additional adjoint equations. The situation is dual to that obtained when optimizing based on direct problems. In usual optimization techniques, in fact, being the shape the control, geometric constraints are easily imposed. The trade-off between usual optimization approaches based on direct problems and our method depends on the number of flow constraints compared to the geometric constraints. In application where the flow quality is vital the present method might be more efficient. In the following, we present a preliminary investigation relative to the application of such method to the flow design of ducted propfan bladings, based on a simplified model.
The blades of the propfan are modeled as flow surfaces of zero thickness which exert forces on the fluid flow. This approximation introduces volume forces in the compressible Euler equations, which is the model adopted for the flow. In our approach, instead of modifying the shape of the flow surfaces modeling the blades, we give the force which the blades exert on the flow, and let the geometry accommodate such distribution of forces. Then the volume force distribution itself varies based on the functional gradient, so that, thrust is maximized. In this approach the blades are represented by those regions where the volume force is non-null. The boundaries of such regions are fixed in the present work, but in general may be themselves susceptible of optimization, as chord distribution and blade tip shape are among the important factors controlling performance.

Flow Model

The flow deflection through rotor of a propfan is the result of the forces that rotor blades exert on the flow. An axial-symmetric model of a turbomachine can be set up by replacing the blades with volume forces.

We assume that the blades have vanishing thickness and infinite solidity, so that the single blade coincides with a stream surface.
Thus, in the case of an inviscid flow, the effect of solid blades is modeled by volume forces orthogonal to stream surfaces. These hypothesis are indeed reasonably verified in usual propfan configurations, where the number of blades is such that solidity is high. Let

$$\mathbf{F} = F^x \mathbf{i} + F^r \boldsymbol{\xi} + F^\vartheta \boldsymbol{\eta} \qquad (1)$$

be the volume force, where $\mathbf{i}, \boldsymbol{\xi}, \boldsymbol{\eta}$ are the unit vectors pertinent to the axial, radial and tangential directions in a cylindrical frame of reference $\{x\mathbf{i}, r\boldsymbol{\xi}, \theta\boldsymbol{\eta}\}$. The distribution of the tangential component $F^\vartheta = F^\vartheta(x, r, \vartheta)$ is the function to be optimized, the same way the shape of a wall is typically optimized in usual optimization algorithms. The geometry of the blades, represented by 2D manifolds

$$\Theta(x, r, \vartheta) = 0 \qquad (2)$$

is found by solving

$$\mathbf{q} - j\omega r\boldsymbol{\eta} \cdot \nabla\Theta = 0 \qquad (3)$$

as the blades are to be stream surfaces of the absolute or relative motion
for stators and rotors respectively.
In the equation above $\mathbf{q} = u\mathbf{i} + w\boldsymbol{\xi} + v\boldsymbol{\eta}$ is the flow velocity vector, ω is the angular velocity of rotors and $j = 0$ for stators, $j = 1$ for rotors.
The components of the volume force F^x and F^r are determined enforcing the blade manifolds to be orthogonal to the volume forces

$$\mathbf{F} \times \nabla\Theta = 0 \qquad (4)$$

which implies

$$F^x = r\frac{\Theta_x}{\Theta_\vartheta}F^\vartheta, \quad F^r = r\frac{\Theta_r}{\Theta_\vartheta}F^\vartheta \qquad (5)$$

Inverse problem

In this section we detail the solution technique of the inverse problem taking as known $F^{\vartheta}(x,r)$
However, it should be noted that this distribution is updated during the optimization in order to maximize a cost function which in our case is thrust.
The solution of the inverse design problem is obtained by means of a time dependent process. The blades can be seen as deformable and impermeable surfaces constrained to the leading edge, like fastened sails waving under the wind effect. An initial configuration of such surface is guessed. The following transient is described by integrating in time the equations governing the time dependent flow motion.
At the end of the transient the blades assume the shape that solve the inverse problem.
In a cylindrical frame of reference, the compressible Euler equations with volume forces acting on the fluid are

$$\mathbf{U}_t + \mathbf{A}_t + \mathbf{B}_t + \mathbf{Q} = 0 \qquad (6)$$

where

$$\begin{aligned}
&\mathbf{U} = \{\rho, u, v, w, e\}^T \\
&\mathbf{A} = \{\rho u,\ p + \rho u^2,\ \rho uv,\ \rho uw,\ u(e+p)\}^T \\
&\mathbf{B} = \{\rho w,\ \rho uw,\ \rho vw,\ p + \rho w^2,\ w(e+p)\}^T \\
&\mathbf{Q} = \{\ \ \rho w/r + \rho u\alpha, \\
&\quad \rho uw/r - F^x + \rho u^2\alpha, \\
&\quad 2\rho vw/r - F^{\vartheta}, \\
&\quad \rho(v^2 - w^2)/r - F^r, \\
&\quad w(p+e)/r - \mathbf{F}\cdot\mathbf{q} + u(p+e)\alpha\}^T
\end{aligned} \qquad (7)$$

as usual ρ is density, p pressure, e total internal energy per unit volume. The boundary condition at the entry section are the two ratios between the velocity components, the total pressure and the total temperature when the flow is subsonic, while all the flow properties are prescribed if the flow is supersonic; at the exit section the static pressure is prescribed if the flow is subsonic, while no boundary conditions are needed when the flow is supersonic. The blade blockage is taken into account by the terms containing the coefficient α, with

$$\alpha = \frac{\partial}{\partial x}[\log(2\pi r - \chi)] \qquad (8)$$

where $\chi = \chi(x,r)$ is the sum of the estimated blade thickness, including the boundary layers.
The system of eqs. (6) is integrated in time using a finite volume formulation based on an approximate Riemann solver to compute the fluxes at cell interfaces with second order spatial accuracy.
A blade surface changes its shape during the transient to obey the condition of impermeability. Let us express eq.(2) as

$$\Theta(x,r,\vartheta,t) = \vartheta - g(x,r,t) = 0 \qquad (9)$$

so that eqs.(5) become

$$F^x = -rg_x F^{\vartheta}, \qquad F^r = -rg_r F^{\vartheta} \qquad (10)$$

Flow particles on $\Theta(x,r,\vartheta,t) = 0$ must remain on the manifold for the impermeability condition.
It follows that during the transient the Langragian derivative of the function $\Theta(x,r,\vartheta,t)$ has to be null

$$\frac{d\Theta}{dt} = \Theta_t + (\mathbf{q} - j\omega r\boldsymbol{\eta})\cdot\nabla\Theta = 0 \qquad (11)$$

that can be written as

$$g_t = -ug_x - wg_r + v/r - j\omega \qquad (12)$$

The above equation is solved coupled to the Euler equations, and it is integrated in time upwinding the spatial derivatives of g according to u and w.

Flow equations adjoint

In this section we derive the functional differential using the ad-

joint technique. As a functional we consider the conventional thrust expressed as

$$T=\left[\int_{r_{hub}}^{r_{tip}}(p+\rho u^2)rdr\right]_{outlet} -\left[\int_{r_{hub}}^{r_{tip}}(p+\rho u^2)rdr\right]_{inlet}=\int_{\Gamma_{io}}H(U)d\Gamma \quad (13)$$

where F^{ϑ}, the distribution of tangential forces, is the control, r_{tip} and r_{hub} are the tip and hub radius respectively.

The following technique is general and can be adapted to other functionals.

The maximum of T is constrained by the steady state Euler equations

$$\mathbf{E}(F^{\vartheta})=\mathbf{A}_x+\mathbf{B}_r+\mathbf{Q}=0 \quad (14)$$

and by the kinematic constraint on the blades

$$g_t(\mathbf{U})=ug_x+wg_r-v/r+j\omega=0 \quad (15)$$

In order to solve such constrained maximization problem we introduce the Lagrangian function

$$\mathfrak{L}(\mathbf{U},g,F^{\vartheta},\Lambda,\mu)=\int_{\Gamma_{io}}H(\mathbf{U})d\Gamma+\int_{\Omega}{}^t\mathbf{\Lambda}\mathbf{E}(\mathbf{U},F^{\vartheta}.g)d\Omega+\int_{\Omega_b}\mu G(\mathbf{U}.g)d\Omega_b \quad (16)$$

where ${}^t\mathbf{\Lambda}(x,r)=(\lambda_1,\lambda_2,\lambda_3,\lambda_4,\lambda_5)$ and $\mu=\mu(x,r)$ are Lagrange multipliers.

A stationary point is found when the variation of $\mathfrak{L}$ with respect to all of its arguments, considered as independent functions, is 0. We have

$$\delta\mathfrak{L}=\delta\mathfrak{L}_{\mathbf{U}}+\delta\mathfrak{L}_{F^{\vartheta}}+\delta\mathfrak{L}_g+\delta\mathfrak{L}_{\mathbf{\Lambda}}+\delta\mathfrak{L}_{\mu} \quad (17)$$

In order to have $\delta\mathfrak{L}=0$, all the single contributions to $\delta\mathfrak{L}$ must vanish at the maximum, so that we enforce

$$\delta\mathfrak{L}_{\mathbf{U}}=0,\quad \delta\mathfrak{L}_g=0$$

$$\delta\mathfrak{L}_{\Lambda}=0,\quad \delta\mathfrak{L}_{\mu}=0 \quad (18)$$

In general this results in $\delta\mathfrak{L}_{F^{\vartheta}}\neq 0$

To reach the maximum we take δF^{ϑ} such that $\delta\mathfrak{L}=\delta\mathfrak{L}_{F^{\vartheta}}>0$, for example using a conjugate gradient method, as explained in the following.

From the condition $\delta\mathfrak{L}_{\mathbf{U}}=0$ we obtain the so called *adjoint* of the Euler equations and its boundary conditions, that is

$${}^t\mathbf{\Lambda}\mathbf{A}_{\mathbf{U}}+{}^t\mathbf{\Lambda}\mathbf{B}_{\mathbf{U}}-{}^t\mathbf{\Lambda}\mathbf{Q}_{\mathbf{U}}-\mu G_{\mathbf{U}}=0 \quad (19)$$

on the domain Ω and

$$\left[H^*_{\mathbf{U}}+{}^t\mathbf{\Lambda}(\mathbf{A}_{\mathbf{U}}n_x+\mathbf{B}_{\mathbf{U}}n_r)\right]\delta\mathbf{U}=0 \quad (20)$$

on its boundary Σ with unit normal vector $\mathbf{n}=(n_x,n_r)$. The function H^* is defined as $H^*=H$ at the inlet and the outlet, and $H^*=0$ elsewhere. The condition $\delta\mathfrak{L}_g=0$ yields to the adjoint of the kinematic constraint

$$(\mu u)_x+(\mu w)_r+\nabla\cdot({}^t\mathbf{\Lambda}\mathbf{K})=0 \quad (21)$$

where

$$K=\mathbf{r}\mathbf{F}^{\vartheta}\begin{Bmatrix}0&0\\1&0\\0&0\\0&1\\u&w\end{Bmatrix}$$

together with the boundary condition

$$[\mu(\mathbf{q}\cdot\mathbf{n})+({}^t\mathbf{\Lambda}\mathbf{K})\cdot\mathbf{n}]\delta g=0 \quad (22)$$

The adjoint equation of the kinematic constraint is coupled to eq.(19) the same way the kinematic constraint is coupled to the flow equations.

Note that the variations of $\mathfrak{L}$ with respect to the Lagrange multipliers $\mathbf{\Lambda}$ and μ simply yield the flow

equations and the kinematic constraint respectively.
Finally, we are left with

$$\delta\Im = \delta\Im_{F^\vartheta} = \int_{\Omega_b} \frac{\partial \mathbf{Q}}{\partial \mathrm{F}^\vartheta} \delta \mathrm{F}^\vartheta d\Omega \qquad (23)$$

This functional depends on $\mathbf{U}, \mathbf{\Lambda}, \mu$ variables that satisfy the flow equations, the kinematic constraint and the respective adjoints. Therefore if we update the present distribution of F^ϑ with

$$\delta \mathrm{F}^\vartheta = \zeta \, {}^t\mathbf{\Lambda} \frac{\partial \mathbf{Q}}{\partial \mathrm{F}^\vartheta} \qquad (24)$$

taking $\zeta > 0$, then $\delta\Im > 0$. Iterating such procedure the maximum is eventually reached. This method, namely the gradient method, has a very slow convergence rate. Better convergence rates are obtained with the conjugate gradient method [6].

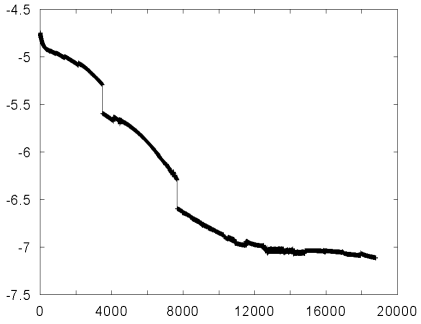


Figure 1. Ducted propfan: gradient residual.

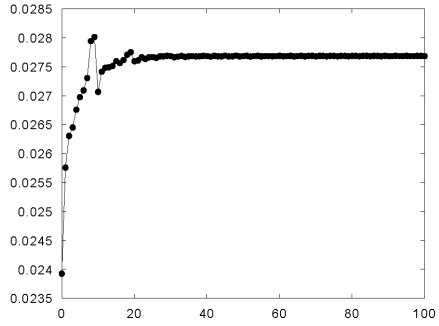


Figure 2. Ducted propfan: Thrust versus optimization steps.

Numerical Results

The distributed control F^ϑ is null everywhere except on the blades, where it is discretized only along the radial direction. We have

$$\mathrm{F}^\vartheta = \☞(r_i)\left[1 - \cos\left(2\pi \frac{x - x_{le}}{x_{te} - x_{le}}\right)\right] \qquad (25)$$

so that the load on the leading x_{le} edge and trailing x_{te} edge is 0.
For each blade considered we have as many design parameters $\☞(r_i)$ as the number of computational points in the radial direction.
Equation (23) is discretized as

$$\delta\Im = \sum_i \delta\☞(r_i) L(r_i)(r_i - r_{i-1}) \qquad (26)$$

where

$$\mathrm{L}(\mathrm{r_i}) = \sum_j {}^t\mathbf{\Lambda}(x_j, r_i) \frac{\partial \mathbf{Q}}{\partial \mathrm{F}^\vartheta}(x_j, r_i) \cdot \left[1 - \cos\left(2\pi \frac{x - x_{le}}{x_{te} - x_{le}}\right)\right](x_j - x_{j-1}) \qquad (27)$$

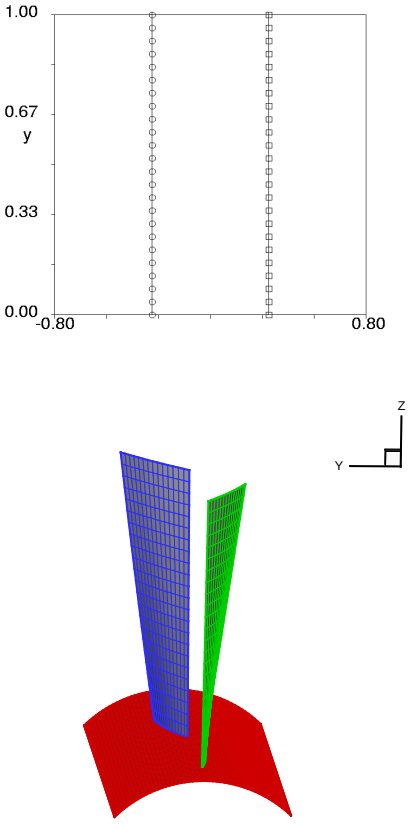


Figure 3. Ducted propfan: initial radial force distribution (top) and blade geometry (bottom) .

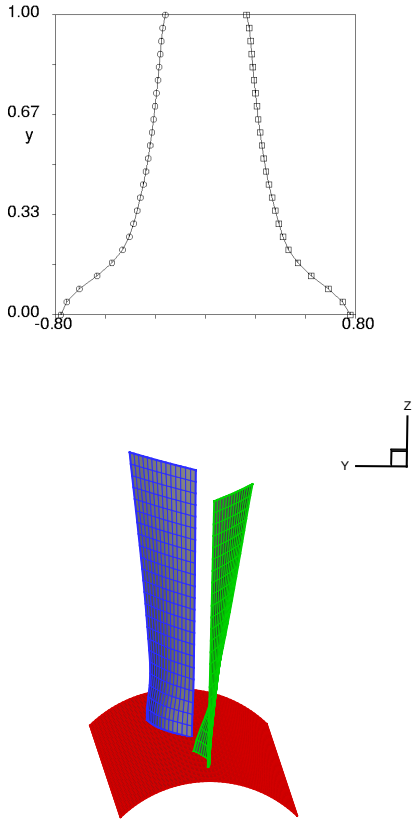


Figure 4.Ducted propfan: fianl radial force distribution (top) and final blade geometry (bottom) .

The gradient method, for example, is obtained taking

$$\delta\vec{F}(r_i) = \zeta L(r_i)\,,\ \zeta > 0\,,$$

so to let $\delta\Theta > 0$ in the discretization accuracy. For the applications, however, we used the conjugate gradient method.
As already mentioned, the formulation proposed in this paper allows an easy treatment of the flow constraints.
When optimizing the fan stage for thrust, it is necessary to keep constant the work done by unit volume by the rotor, that is

$$\int_{\Omega_b} F^{\vartheta}\omega r^2 dr = \text{constant} \qquad (28)$$

where ω is the angular speed.
The increments $\delta\vec{F}(r_i)$ are constrainned to lay on the manifold determined by eq. (28).
Discretizing and linearizing such equation, we obtain an hyperplane onto which the gradient can be projected in order to satisfy the constraint.
This procedure is much less expensive and complicated than solving an additional partial differential equation (PDE) for the constraint as it would be necessary for usual optimization methods.
Clearly the trade-off should be critically evaluated in relation to eventual geometric constraint, which in our method would be treated by an additional PDE.
In this preliminary study the region where the force is non-null is fixed in the meridional plane.

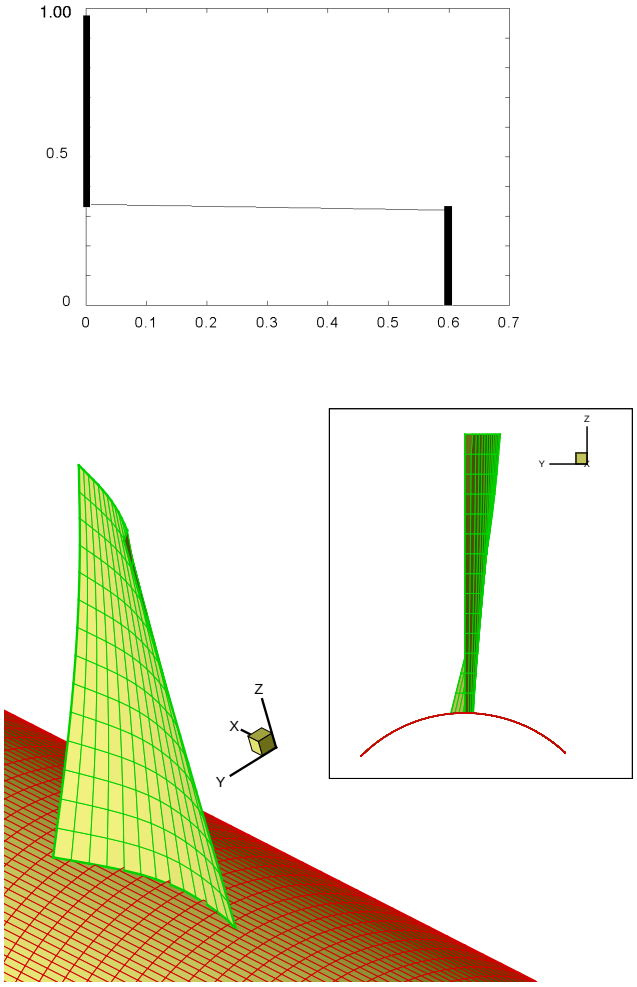


Figure 5. Unducted propfan: initial radial force distribution (top) and blade geometry (bottom).

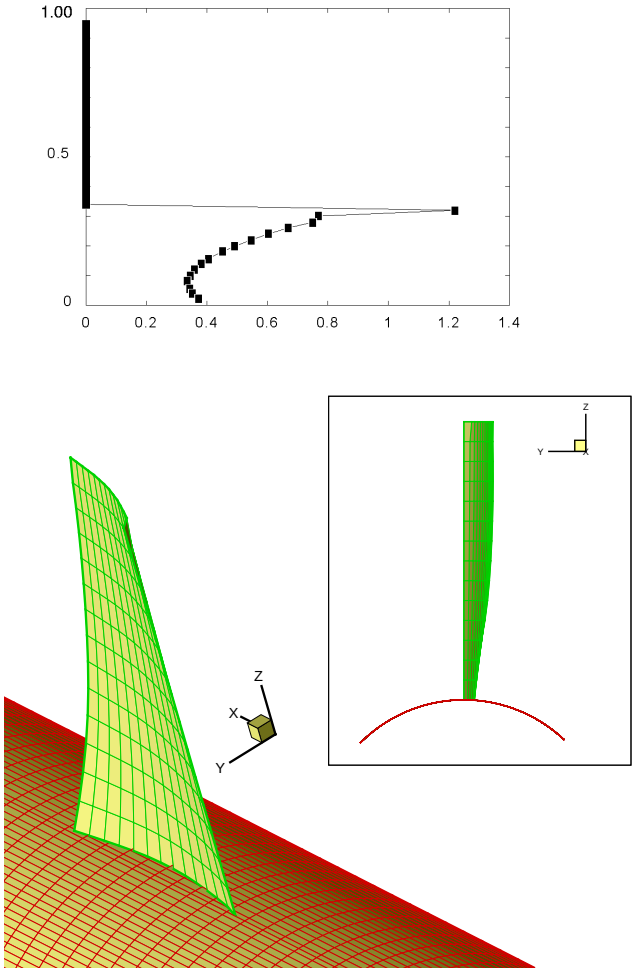


Figure 6. Unducted propfan: final radial force distributon (top) and blade geometry (bottom).

Indeed, additional investigation is now devoted to a different force parameterization making feasible to optimize the blade planform, besides the force intensity.
Also, other functionals may be more significant than thrust for propfan blades optimization.
Noise emission, for example, is known to be related to blades load.
However in this first step we limited our investigation to thrust optimization test cases.

As first example, the optimization of a ducted propfan configuration is presented. The system consist in two counter-rotating rotors. The design variables for this test case, being the grid 75×25, are 25 for the first rotor and 25 for second rotor; $\omega = 0.4$ for the first rotor and $\omega = -0.4$ for the second. The initial force distribution is constant in the radial direction for both rotors, and is shown in figure 3, togheter with the initial blade geometry. In figure 4 the force distribution and blade geometry obtained after optimization are shown. Indeed the thrust gradient decreases of more than two orders of magnitude, and thrust increases of 8%, as shown in figs. 1 and 2.

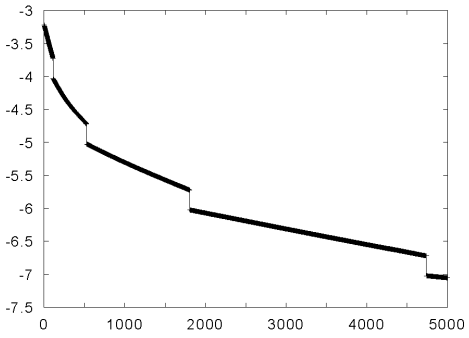


Figure 7.Unducted propfan: gradient residual versus optimization steps.

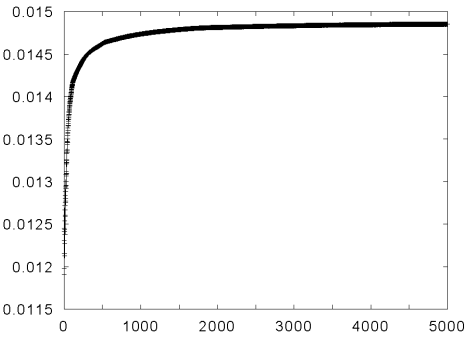


Figure 8. Unducted propfan: thrust versus optimization steps.

The thrust overshoot in the first optimization iterations is explained by the fact that the flow solution is not fully converged.
Flow and adjoint equations convergence is reached when the gradient residual decreases at least of one order of magnitude. Neverthless, it can be seen that, in this case, the maximum thrusth is reached in about 30 optimization iterations.
The second example refers to the optimization of an unducted propfan for thrust at flight Mach number is 0.6. The rotor angular speed is $\omega = 0.4$.

From the comparison of the radial force distributions for the two cases presented (figs. 3,4 and 5,6) we can conclude that, maximizing thrust, the ducted configuration is close to free vortex design, while the unducted case tends towards higher tip-loads, according to Betz's optimum condition (see Ref. [7]).

Conclusion

In this work we propose and derive an hybrid method for aerodynamic design, and apply it to propfan design. It takes advantage of the inverse solution of the equations to determine optimal flows.
The preliminary applications shown are a first step towards optimization of the blade planform and load for noise reduction.

References

[1] Ferlauto M., Iollo A., Zannetti L., (2000). "Coupling of Optimization and Inverse Problem for Aerodynamic Shape Design", AIAA paper 2000-0668.

[2] Iollo A., Salas M.D.,(1996). "Contribution to the optimal shape design of 2D internal flows with embedded shocks", J. of Comp.Phys. 125, 124-134.

[3] Zannetti L.(1980)."Time dependent method to solve the inverse problem for internal flows", AIAA J., 18, 754--758.

[4] Zannetti L., Larocca F. (1990). "Inverse Methods for 3D Internal Flows", AGARD-R-780.

[5] Bena C., Larocca F., Zannetti L. (1999). "Design of multistage axial flow turbines and compressors", Proc. IMech-E 3rd Conference on Turbomachinery, p. 635-644, London.

[6] Fletcher R., "Practical methods of Optimization", John Wiley & Sons, New York, (1980)

[7] Glauert F.,"Airplane propellers" in "Aerodynamic Theory", vol.4, pp 169-360, Durand editor, J. Verlag, (19350)